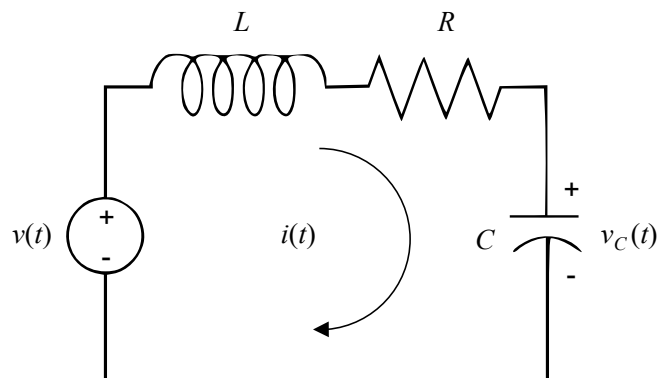




Advanced Control System Design - Tutorial 1 – Solution

1. Find the transfer function that relates the capacitor voltage to the input voltage in the following circuit.



Solution:

$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int_0^t i(\tau) d\tau = v(t)$$

$$i(t) = \frac{dq(t)}{dt}$$

$$L \frac{d^2q(t)}{dt^2} + R \frac{dq(t)}{dt} + \frac{1}{C} q(t) = v(t)$$

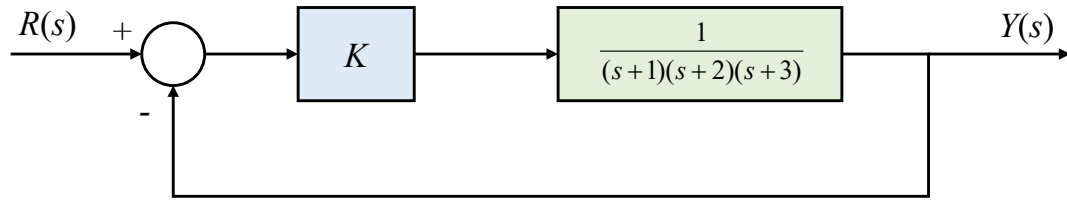
$$q(t) = Cv_C(t)$$

$$LC \frac{d^2v_C(t)}{dt^2} + RC \frac{dv_C(t)}{dt} + v_C(t) = v(t)$$

$$(LCs^2 + RCs + 1)V_C(s) = V(s)$$

$$\frac{V_C(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

2. Investigate the open-loop and closed-loop stabilities for the system shown below, when $K = 10$ and 200 :



Solution:

When $K = 10$:

$$OLTF = \frac{K}{(s+1)(s+2)(s+3)} = \frac{10}{(s+1)(s+2)(s+3)} \Rightarrow \text{The response is stable}$$

$$CLTF = \frac{K}{(s+1)(s+2)(s+3) + K} = \frac{10}{s^3 + 6s^2 + 11s + 16}$$

$$CLTF = \frac{10}{(s+4.31)(s+0.85+j1.73)(s+0.85-j1.73)} \Rightarrow \text{The response is stable}$$

When $K = 200$:

$$OLTF = \frac{K}{(s+1)(s+2)(s+3)} = \frac{200}{(s+1)(s+2)(s+3)} \Rightarrow \text{The response is stable}$$

$$CLTF = \frac{K}{(s+1)(s+2)(s+3) + K} = \frac{200}{s^3 + 6s^2 + 11s + 206}$$

$$CLTF = \frac{200}{(s+7.91)(s-0.95+j5.02)(s-0.95-j5.02)} \Rightarrow \text{The response is unstable}$$

3. Find the determinant of the following matrix:

$$\mathbf{N}_1 = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 2 & 9 & 7 \end{bmatrix} : |\mathbf{N}_1| = 18$$

4. Find the inverse of the following matrices:

$$\mathbf{O}_1 = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} : \mathbf{O}_1^{-1} = \frac{1}{|\mathbf{O}_1|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -0.5 & 0.5 \end{bmatrix}$$

$$\mathbf{O}_2 = \begin{bmatrix} 1 & 2 & 2 \\ 4 & 6 & 4 \\ 2 & 8 & 4 \end{bmatrix} :$$

$$\mathbf{O}_2^{-1} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^{-1} = \frac{1}{|\mathbf{O}_2|} \begin{bmatrix} + \begin{vmatrix} e & f \\ h & i \end{vmatrix} & - \begin{vmatrix} d & f \\ g & i \end{vmatrix} & + \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ - \begin{vmatrix} b & c \\ h & i \end{vmatrix} & + \begin{vmatrix} a & c \\ g & i \end{vmatrix} & - \begin{vmatrix} a & b \\ g & h \end{vmatrix} \\ + \begin{vmatrix} b & c \\ e & f \end{vmatrix} & - \begin{vmatrix} a & c \\ d & f \end{vmatrix} & + \begin{vmatrix} a & b \\ d & e \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} -0.5 & 0.5 & -0.25 \\ -0.5 & 0 & 0.25 \\ 1.25 & -0.25 & -0.125 \end{bmatrix}$$

5. Find the rank of the following matrices:

$$\mathbf{P}_1 = \begin{bmatrix} \boxed{1} & \boxed{0} & 0 & 0 \\ \boxed{0} & \boxed{1} & 0 & 0 \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is 1, and the rank is 2.}$$

$$\mathbf{P}_2 = \begin{bmatrix} \boxed{1} & 0 & \boxed{0} & 0 \\ 0 & 0 & \boxed{1} & 0 \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is 1, and the rank is 2.}$$

$$\mathbf{P}_3 = \begin{bmatrix} \boxed{1} & 0 & 0 & \boxed{0} \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is 1, and the rank is 2.}$$

$$\mathbf{P}_4 = \begin{bmatrix} \boxed{0} & 0 & 0 & \boxed{1} \\ 1 & 0 & 0 & \boxed{0} \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is -1, and the rank is 2.}$$

$$\mathbf{P}_5 = \begin{bmatrix} \boxed{0} & 0 & \boxed{2} & \boxed{1} \\ 1 & 0 & \boxed{0} & \boxed{0} \end{bmatrix} \Rightarrow \text{The determinants of the sub-matrices are 0, -1 and -2, and the rank is 2.}$$

$$\mathbf{P}_6 = \begin{bmatrix} \boxed{1} & 0 & 0 & \boxed{2} \\ 3 & 0 & 0 & \boxed{6} \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is 0, and the rank is 1.}$$

$$\mathbf{P}_7 = \begin{bmatrix} \boxed{1} & \boxed{2} & \boxed{3} & 0 \\ 1 & 8 & 7 & 0 \\ 4 & 5 & 6 & 0 \end{bmatrix} \Rightarrow \text{The determinant of the sub-matrix is -24, and the rank is 3.}$$

$$\mathbf{P}_8 = \begin{bmatrix} \boxed{1} & 0 & \boxed{3} & 0 \\ 0 & 0 & 0 & 0 \\ \boxed{4} & 0 & \boxed{6} & 0 \end{bmatrix} \Rightarrow \text{Since there is one row with all zeros, then the maximum rank can be 2}$$

or less. The determinant of the sub-matrix is -6, and the rank is 2.

$$\mathbf{P}_9 = \begin{bmatrix} \boxed{1} & 0 & \boxed{3} & 0 \\ 0 & 0 & 0 & 0 \\ \boxed{2} & 0 & \boxed{6} & 0 \end{bmatrix} \Rightarrow \text{Since there is one row with all zeros, then the maximum rank can be 2}$$

or less. The determinant of the sub-matrix is 0, and the rank is 1.

$$\mathbf{P}_{10} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{Since there are two rows with all zeros, then the rank is 1.}$$

$$\mathbf{P}_{11} = \begin{bmatrix} 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{Since there are two rows with all zeros, then the rank is 1.}$$

$$\mathbf{P}_{12} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{Since all the elements are zeros, then the rank is 0.}$$